Supplement: extended methods for scoped generalization

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Supplement: extended methods for scoped generalization

This supplement documents three method extensions that broaden the toolkit without changing the categorical federated claims of the main text.

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Supplement: extended... (part 2)

Each answers a “does it generalize?” question raised by a specific main-text result and each connects back to it.

Supplement: extended... (part 3)

The additional contamination models, gallery, and onset sweep stress-test the robustness verdict of sec. 10.5 beyond its single confident-wrong mechanism; the Gaussian divergence bridge points toward the continuous-state direction of sec. 11.22;

Supplement: extended... (part 4)

and greedy multi-hypothesis reduction extends the configured BMR sign control of sec. 10.4 from one pruned state to a family.

Supplement: extended... (part 5)

Each is tested and isolated; none participates in the headline robustness verdict, so the main-text claims stand or fall without them.

Continuous-state divergence bridge for Gaussian beliefs

Every robustness claim in the main text is discrete-categorical, matching Friston's worked example.

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Continuous-state divergence… (part 2)

To show the divergence family carries over to the Gaussian beliefs a continuous-state active-inference extension would use, `divergences.py` adds closed forms for 1-D Gaussians: the Kullback–Leibler divergence

Continuous-state divergence... (part 3)

$$\begin{aligned}&\mathrm{KL}(\mathcal{N}(\mu_q, \sigma_q^2) \parallel \mathcal{N}(\mu_p, \sigma_p^2)) \\ &= \frac{1}{2}\left[\begin{array}{c}\frac{\sigma_q^2}{\sigma_p^2} + \frac{(\mu_p - \mu_q)^2}{\sigma_p^2} \\ -1 + \log\frac{\sigma_p^2}{\sigma_q^2}\end{array}\right].\end{aligned} \tag{32}$$

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Continuous-state divergence... (part 4)

and the α-Rényi divergence with interpolated variance $\sigma_\alpha^2 = \alpha\sigma_p^2 + (1-\alpha)\sigma_q^2$,

Continuous-state divergence... (part 5)

$$
\begin{aligned}
&D_\alpha(\mathcal{N}_q \,\|\, \mathcal{N}_p) \\
&\quad = \frac{\alpha(\mu_q - \mu_p)^2}{2\sigma_\alpha^2} \\
&\qquad - \frac{1}{2(\alpha-1)} \log \frac{\sigma_\alpha^2}{\sigma_q^{2(1-\alpha)} \sigma_p^{2\alpha}}.
\end{aligned}
\tag{33}
$$

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Continuous-state divergence... (part 6)

As in the categorical case (eq. 6 and Lemma 3), eq. 33 recovers eq. 32 in the $\alpha \to 1$ limit, and the closed form is returned exactly inside a small band around one. These functions are **out of scope** for the federated experiments —

Continuous-state divergence... (part 7)

they are wired into no aggregation rule or sweep — and exist purely as the explicitly-scoped bridge toward continuous active inference.

Additional contamination models for the robustness surface

`contamination.py` extends the confident-wrong, label-noise, and uniform models with two mechanisms that probe different attack surfaces, both honoring the identity anchor (rate zero returns the belief unchanged):

Additional contamination... (part 2)

Byzantine targeted. A *multiplicative* log-odds tilt toward an adversary-chosen state, $s' \propto s \cdot \exp(\text{rate} \cdot \text{tilt} \cdot e_{\text{target}})$. Unlike the additive convex mixes,

Additional contamination… (part 3)

the corruption compounds with the belief's own shape, so the non-target states keep their relative order — the canonical targeted poisoning of a product-of-experts pool.

Additional contamination… (part 4)

Drift. A *slowly-moving* bias grows linearly across communication rounds via a phase $\phi = \text{round}/(\text{rounds} - 1)$, so the first round is clean and the bias creeps in. This is the stealthy sentinel whose miscalibration only becomes confident late, defeating any one-shot screen.

Additional contamination… (part 5)

Both are exercised in the contamination tests; the headline sweep continues to use the confident-wrong model so the verdict is comparable to the main text.

Contamination gallery by corruption mechanism

To check the robust-beats-naive result is not an artifact of the single confident-wrong mechanism — and not of a single lucky seed —

Contamination gallery by… (part 2)

the `run_contamination_gallery` function in the `experiments` module re-runs the paired comparison ($n = 24$ trials across 64 independent seeds, contamination strength 0.60) under every model.

Contamination gallery by... (part 3)

For each mechanism it selects one robust method by pooled mean consensus accuracy **for descriptive gallery display only**, then reports that displayed member's robust-minus-naive difference, 95% seed bootstrap interval, and *win fraction* — the fraction of seeds in which the displayed member beats naive.

Contamination gallery by... (part 4)

This is not the selection-free inferential surface; the main-text all-method review grid in sec. 10.5.1 serves that role.

Contamination gallery by… (part 5)

Mechanism	Class	Naive mean	Display robust mean (preset)
byzantine	directional	0.6306	0.6599 (beta)
confident wrong	directional	0.9836	0.9878 (AR)

Contamination gallery by… (part 6)

Mechanism	Mean robust - naive	95% CI	Win fraction	Reliable
byzantine	0.0293	[0.0124, 0.0472]	0.62	No

Contamination gallery by… (part 7)

The paired contamination summaries tbls. 14, 15 form one descriptive sensitivity screen, deliberately narrower than “robust always wins.” The pooled display member has a positive all-seed and interval pattern under confident wrong, drift —

Contamination gallery by… (part 8)

the *additive* directional attacks.

The full set of directional mechanisms is confident wrong, byzantine, drift; the **byzantine** attack is directional too, but its *multiplicative* log-odds tilt escalates faster:

Contamination gallery by... (part 9)

at this strength it sits near a veto cliff where the naive pool is already badly degraded and the displayed robust advantage does not hold across seeds (its win fraction is well below the 0.95 display bar), so it does not clear the descriptive reliability screen.

Contamination gallery by… (part 10)

The paired mean-difference interval is reported separately in tbl. 15.

Contamination gallery by… (part 11)

The **entropy** attacks (label noise, uniform) raise entropy or inject noise without a fixed wrong target, so the product-of-experts is not pulled off the truth and there is nothing to beat — the robust members stay close rather than winning (naive undegraded by entropy attacks: Yes).

Contamination gallery by... (part 12)

fig. 25 draws all mechanisms with their win fractions. This is the honest scope of this configured gallery: its displayed members separate from naive under the declared *sustained additive* directional contamination, stay close under the declared entropy attacks,

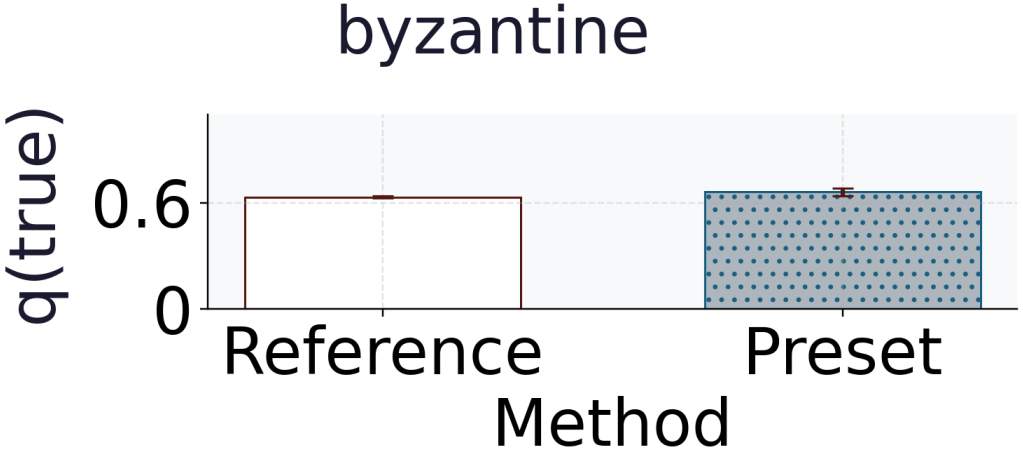
Contamination gallery by... (part 13)

and lose the displayed advantage against the tested multiplicative adversary near the veto regime.

Contamination gallery by… (part 14)

These finite cells do not establish the same ordering for every attack strength or world, and they do not turn a pooled display selection into selection-free inference.

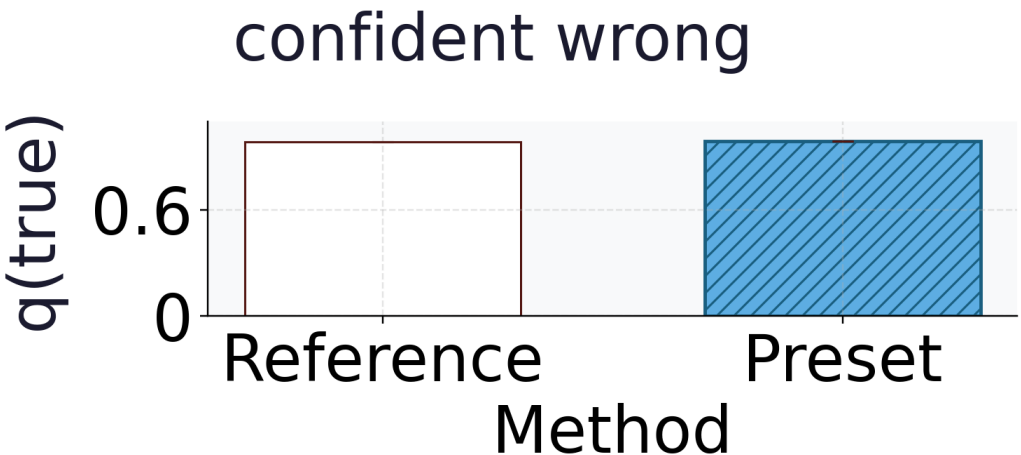
Contamination gallery by… (part 15)



Contamination gallery by… (part 16)

byzantine: selected preset beta; descriptive win fraction 0.62; below bar.

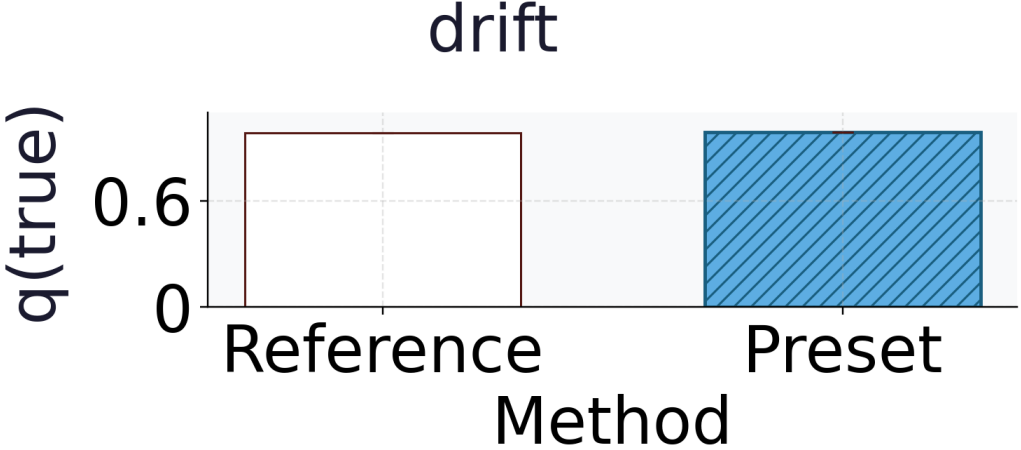
Contamination gallery by… (part 17)



Contamination gallery by… (part 18)

confident wrong: selected preset AR; descriptive win fraction 1.00; flagged.

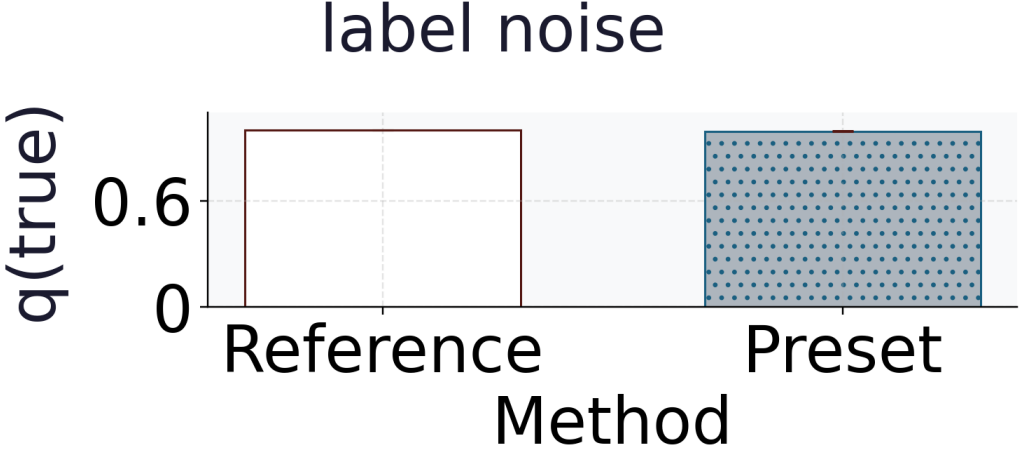
Contamination gallery by… (part 19)



Contamination gallery by… (part 20)

drift: selected preset AR; descriptive win fraction 1.00; flagged.

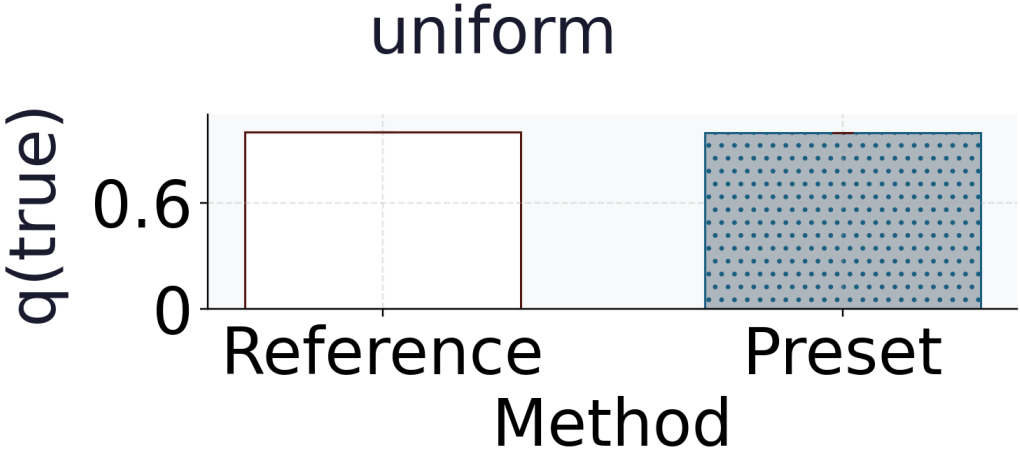
Contamination gallery by… (part 21)



Contamination gallery by… (part 22)

label noise: selected preset AR; descriptive win fraction 0.00; below bar.

Contamination gallery by… (part 23)



Contamination gallery by… (part 24)

uniform: selected preset AR; descriptive win fraction 0.00; below bar.

Contamination gallery by… (part 25)

Open reference bars retain their hatch. Selected presets use full fill when flagged and pale dotted fill below the display bar. Color is not the sole flag encoding.

Contamination gallery by… (part 26)

Each bar has its own 95% seed-bootstrap interval.

Contamination gallery by… (part 27)

Display flags: 2/5. Win fractions are descriptive, not p-values. Selected presets do not define fixed-preset confirmatory contrasts.

Robustness onset by corruption mechanism

The gallery fixes one contamination strength; `experiments.run_robustness_onset` maps the *rate dependence* ($n = 24$ trials $\times$ 64 seeds per rate).

Robustness onset by... (part 2)

For each directional mechanism it reports the **descriptive onset rate** — the smallest rate at which the pooled-selected display member's win fraction reaches 0.95 —

Robustness onset by... (part 3)

and that member's versus naive accuracy at the worst (highest) swept rate. These display summaries are not selection-free inference;

the main-text all-method review grid in sec. 10.5.1 is the inferential surface:

Robustness onset by... (part 4)

Mechanism	Onset rate	Naive @ worst	Robust @ worst	Robust method @ worst
byzantine	0.4	0.0165	0.0000	beta
confident wrong	0.6	0.6676	0.7772	beta

Robustness onset by... (part 5)

The mechanism-specific onset thresholds are collected in tbl. 16.

Robustness onset by… (part 6)

The rate dependence sharpens the gallery's snapshot, and fig. 26 draws it. The additive confident-wrong and drift attacks degrade the naive pool gradually; past their onset rate the robust member stays above it through to the worst rate.

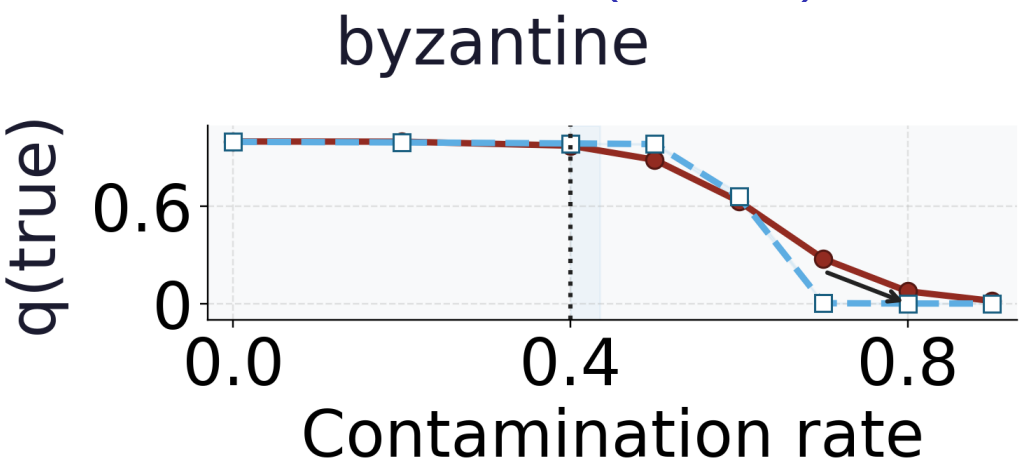
Robustness onset by... (part 7)

The multiplicative byzantine attack is qualitatively different: it opens an *early* robustness window — robust overtakes at a lower onset rate — but then escalates to the veto cliff where naive and robust both collapse, so its worst-rate accuracy is near zero for both.

Robustness onset by... (part 8)

This is the rate-resolved bounded interpretation: the displayed contrast is sustained against the two additive directional mechanisms and only transient against the multiplicative mechanism.

Robustness onset by... (part 9)



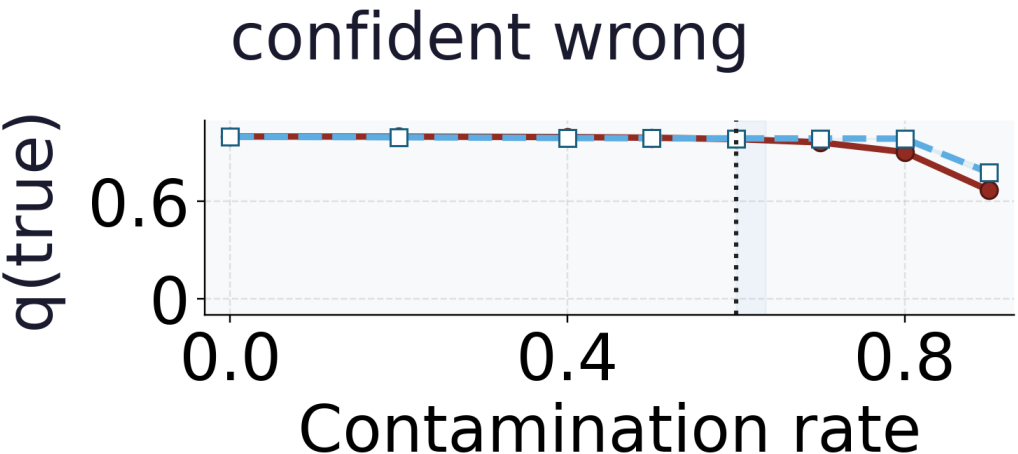
Robustness onset by… (part 10)

byzantine. Onset: 0.4. Gap at maximum rate: -0.016. Each curve has its own 95% seed-bootstrap band.

Robustness onset by… (part 11)

Arrow: declared byzantine veto cliff near the penultimate rate. The onset window does not establish universal robustness.

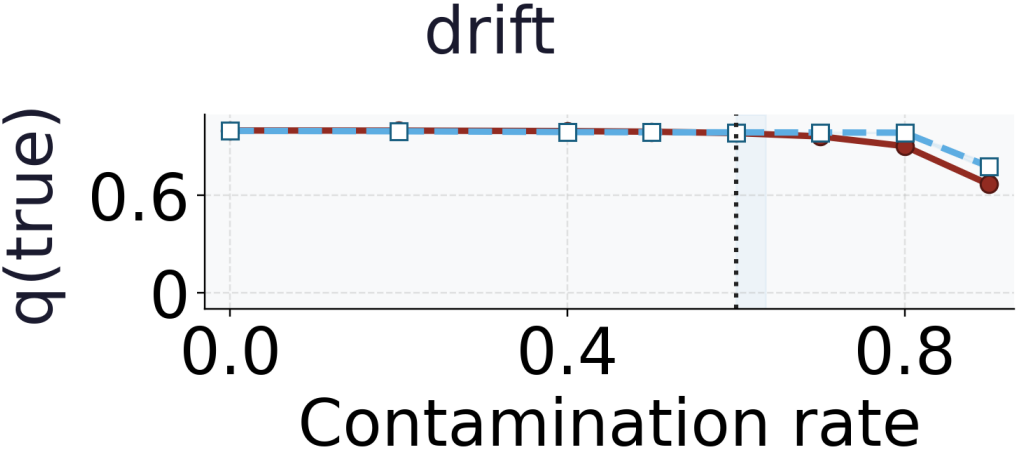
Robustness onset by... (part 12)



Robustness onset by… (part 13)

confident wrong. Onset: 0.6. Gap at maximum rate: +0.110. Each curve has its own 95% seed-bootstrap band.

Robustness onset by... (part 14)



Untagged PDF derivative | HTML reader

Robustness onset by… (part 15)

drift. Onset: 0.6. Gap at maximum rate: +0.110. Each curve has its own 95% seed-bootstrap band.

Robustness onset by… (part 16)

Filled circles / solid: reference log pool. Open squares / dashed: pooled display server preset. This selected display curve is not a fixed-preset confirmatory contrast.

Conditional world and attack-geometry grid

The finite MAJ-1 characterization is now extended across 40 preregistered world/scenario cells: two hidden-state locations, two observability levels, five attack mechanisms,

Conditional world and... (part 2)

and two adversarial weight settings.

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Conditional world and... (part 3)

The independent unit is the seeded world/scenario row; each cell averages 24 nested trials over 64 seeds before the matched contrast is formed. The primary estimand is naive true-state error minus robust true-state error,

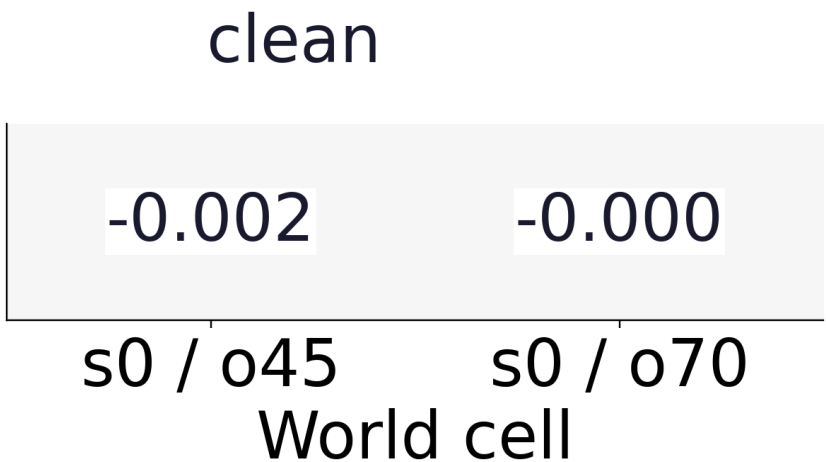
Conditional world and... (part 4)

so a positive value means the robust heuristic assigns more true-state mass in that finite cell.

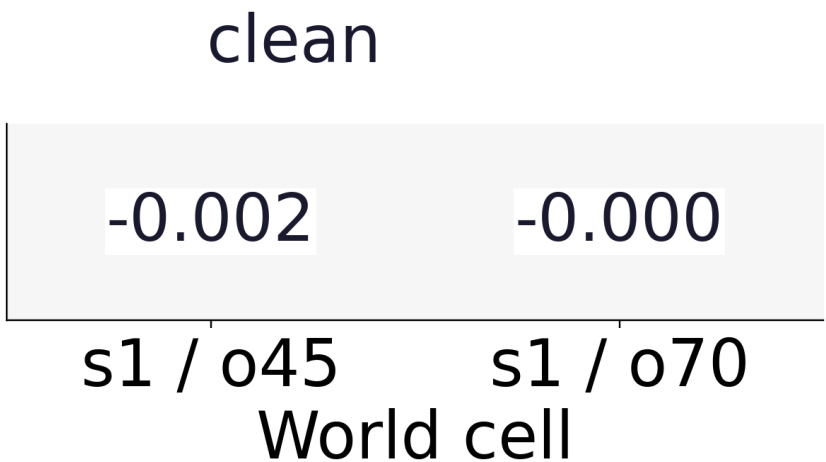
Conditional world and... (part 5)

The robustness-zero control is pass, and the report remains explicitly labelled `conditional_finite_grid`. The resulting conditional surface is shown in fig. 27.

Conditional world and... (part 6)

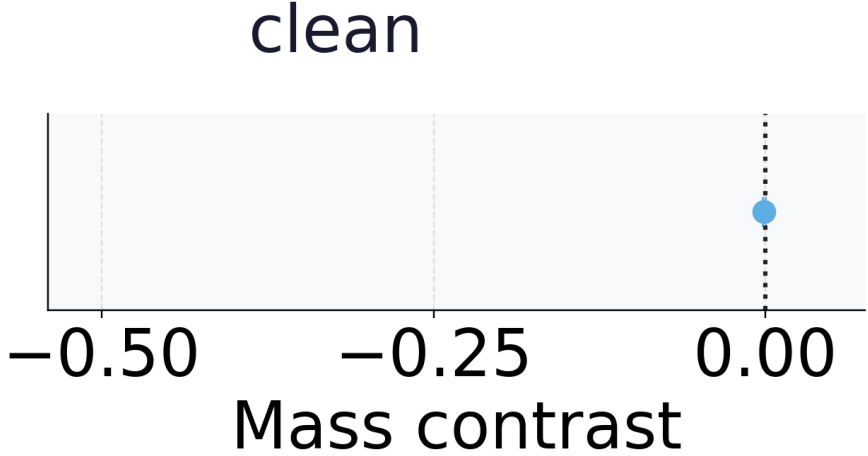


Conditional world and... (part 7)



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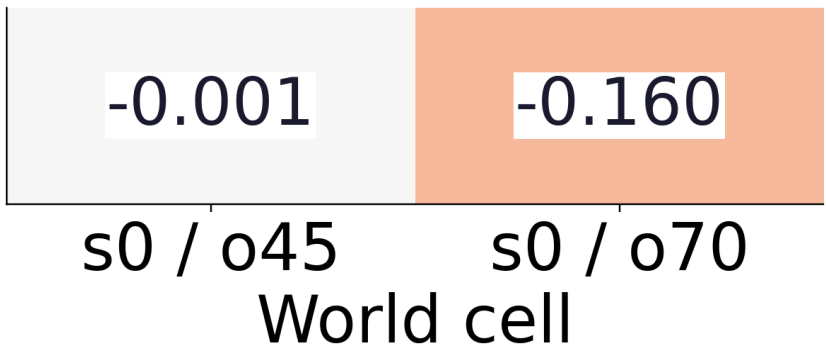
Conditional world and... (part 8)



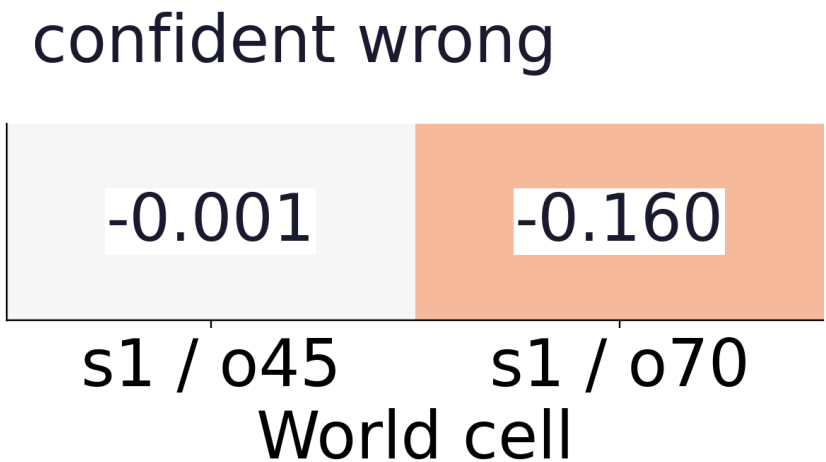
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Conditional world and... (part 9)

confident wrong

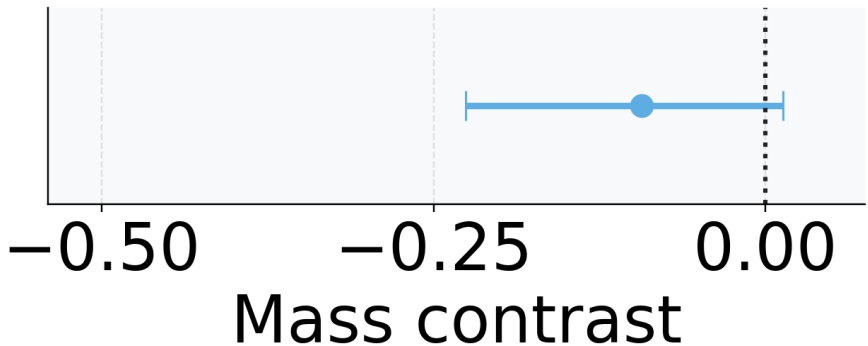


Conditional world and... (part 10)



Conditional world and... (part 11)

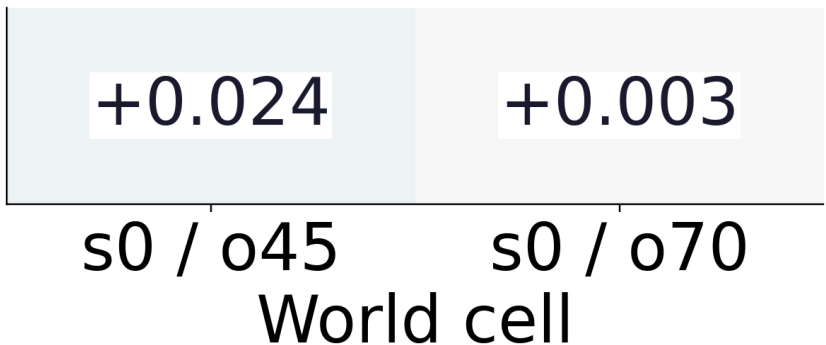
confident wrong



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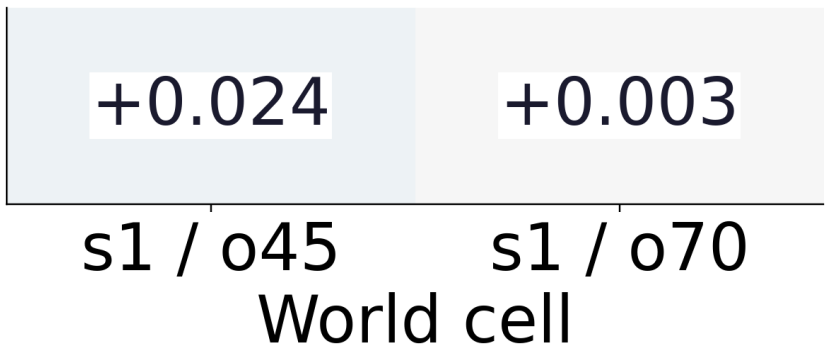
Conditional world and... (part 12)

permutation



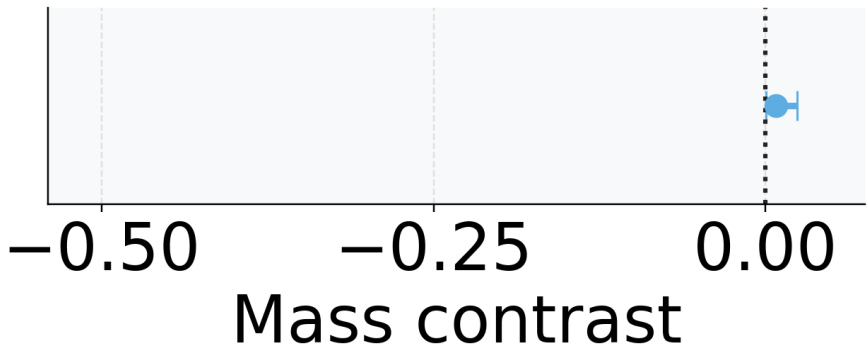
Conditional world and... (part 13)

permutation

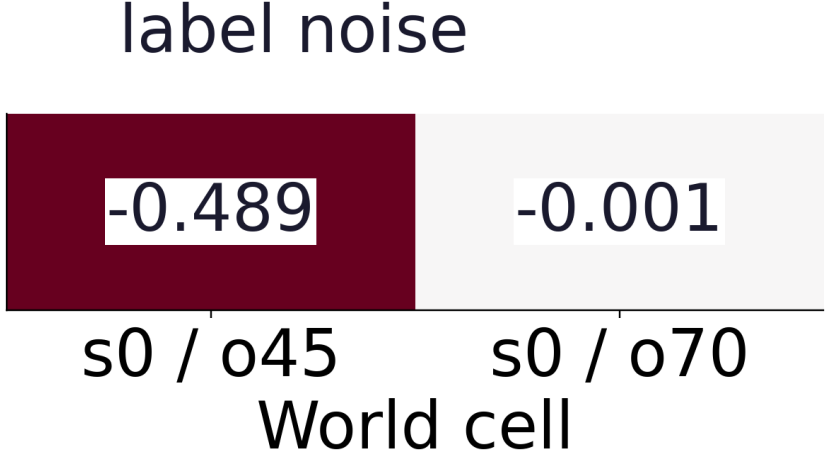


Conditional world and... (part 14)

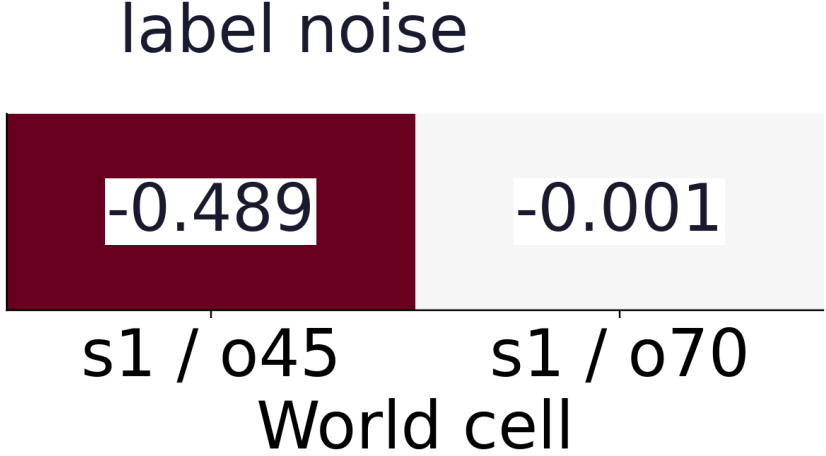
permutation



Conditional world and... (part 15)

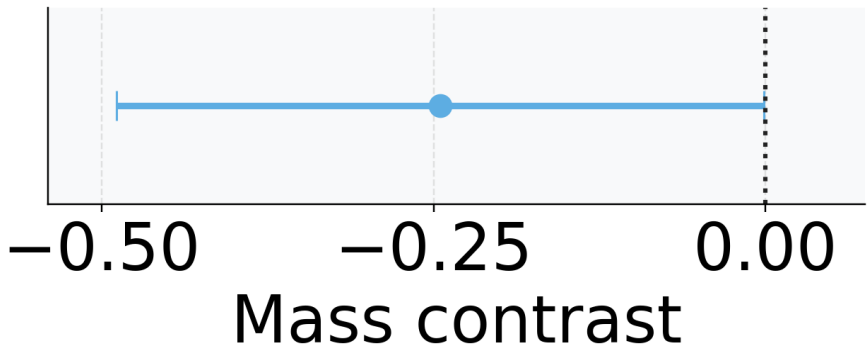


Conditional world and... (part 16)

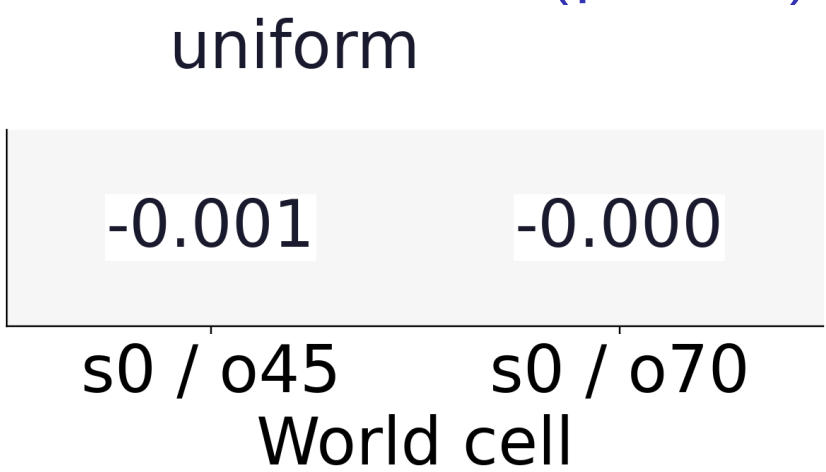


Conditional world and... (part 17)

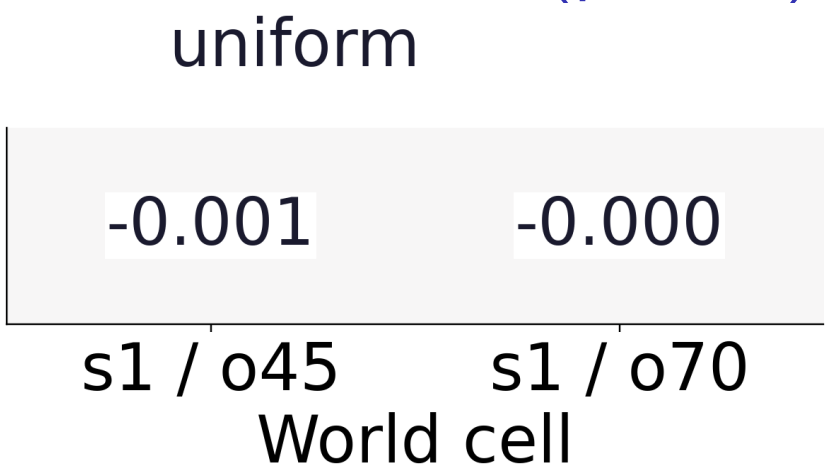
label noise



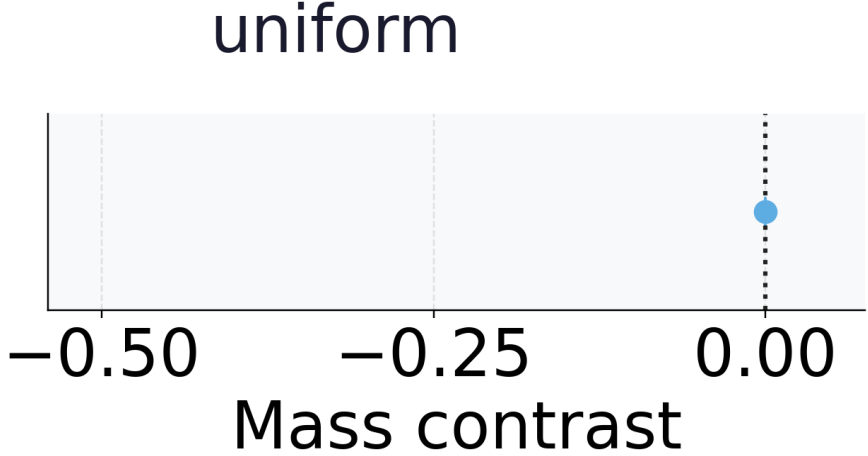
Conditional world and... (part 18)



Conditional world and... (part 19)



Conditional world and... (part 20)



Conditional world and... (part 21)

Seed-level true-state-mass contrast:
naive error minus robust error.
Positive values favor the heuristic
on the declared finite grid.

Conditional world and… (part 22)

Heatmap: adversary weight 1. State s0/s1; observability o45/o70. Common color scale: [-0.4889, 0.4889]. Every cell prints its signed mean.

Conditional world and... (part 23)

Summary circles: finite-grid means across configured cells per attack. Asymmetric capped whiskers reach the observed minimum and maximum; they are not confidence intervals.

Conditional world and... (part 24)

Dark dotted rule: zero method contrast. This finite-grid summary does not support population, deployment or universal robustness claims.

Proper scores and calibration controls

Argmax accuracy does not distinguish a cautious posterior from an overconfident one. The scoring extension therefore pre-registers the paired seed-level categorical log-score difference between naive and robust consensus beliefs as its primary belief-quality estimand.

Proper scores and... (part 2)

The Brier score and equal-width expected calibration error are secondary diagnostics; higher log score and lower Brier/ECE are better. Agents and trials remain nested within the seed.

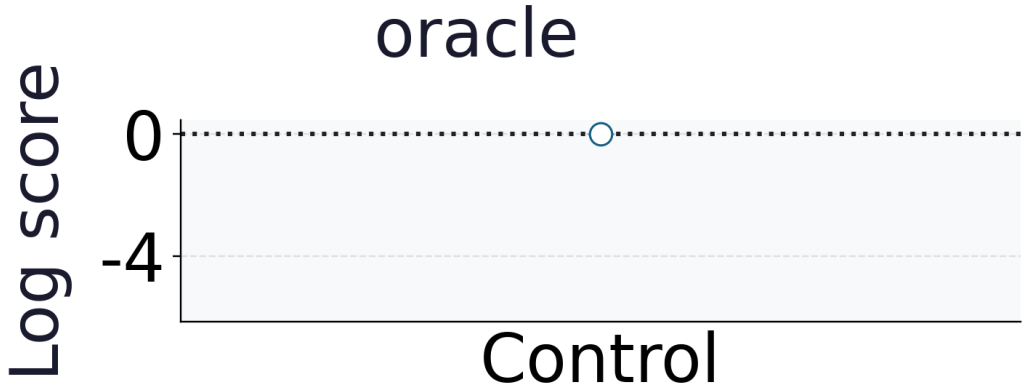
Proper scores and... (part 3)

The report also includes three negative controls: an oracle, a uniform belief, and a confidently wrong belief. Their expected score ordering is checked before any method contrast is interpreted: oracle $>$ uniform $>$ confidently wrong under the clipped log score. The control ordering gate is `pass`,

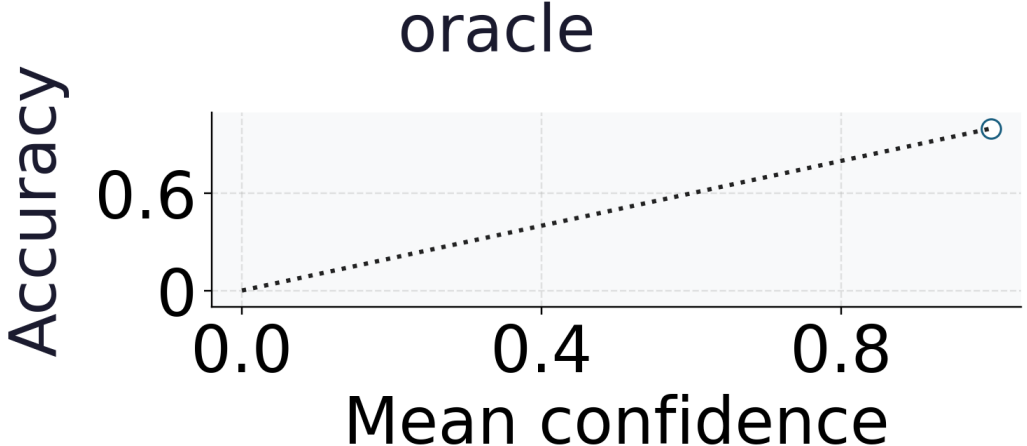
Proper scores and… (part 4)

and the confidently-wrong-versus-uniform control is pass, using 64 independent seeds and 24 nested trials per seed. The diagnostic is shown in fig. 28.

Proper scores and... (part 5)



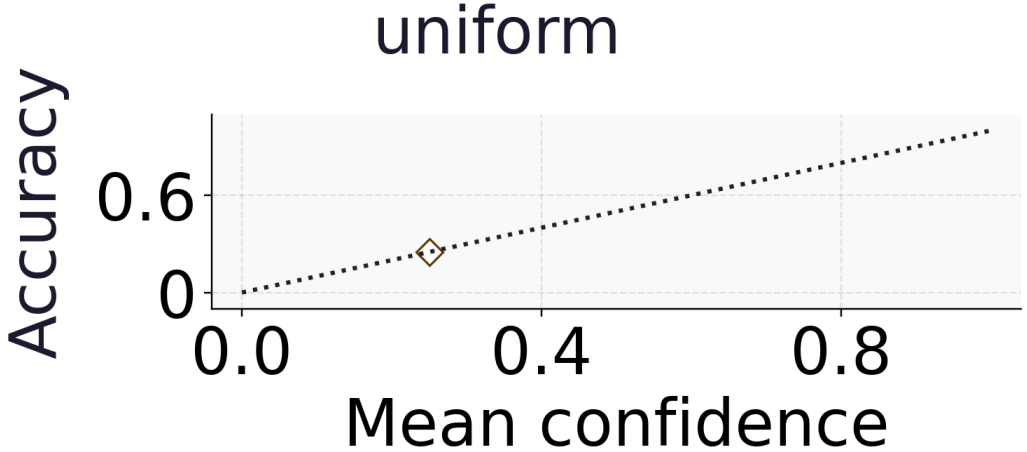
Proper scores and... (part 6)



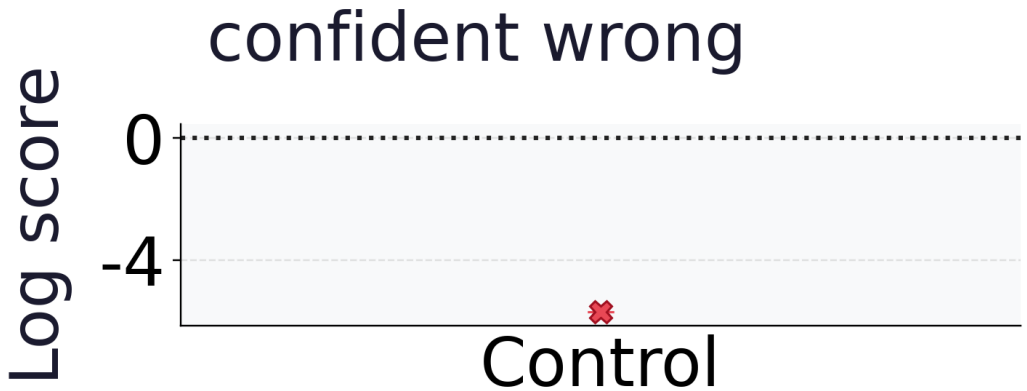
Proper scores and... (part 7)



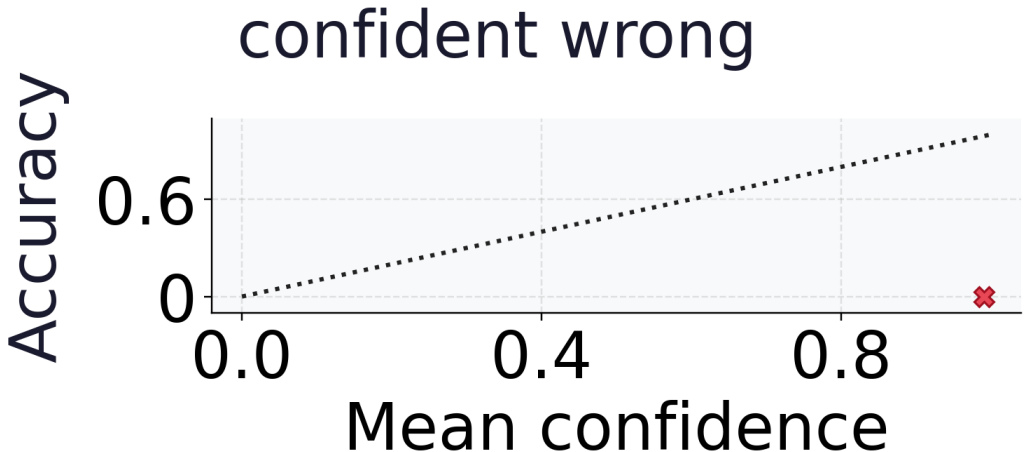
Proper scores and... (part 8)



Proper scores and... (part 9)



Proper scores and... (part 10)



Proper scores and… (part 11)

Categorical log score in nats: higher is better. Whiskers are 95% seed-bootstrap intervals; seed is the independent unit.

Proper scores and… (part 12)

Reliability is a descriptive control diagnostic. Every finite bin is drawn on common 0–1 axes. The dark dotted identity marks perfect calibration.

Proper scores and… (part 13)

Oracle, uniform and confident-wrong controls check score and reliability behavior. They do not establish deployment calibration or universal robustness.

Greedy multi-hypothesis model reduction beyond the main BMR study

The single-step reduction of sec. 8.4 scores one candidate reduced prior. `bayesian_model_reduction.py` adds `greedy_reduce`, which performs structure learning over a *family* of redundant states:

Greedy multi-hypothesis... (part 2)

starting from the full prior, it scores pruning each not-yet-pruned state against the current reduced prior, accepts the single prune with the largest positive free-energy gain,

and repeats until no remaining prune improves model evidence.

Greedy multi-hypothesis... (part 3)

Every accepted step has a strictly positive incremental ΔF, so the cumulative evidence is monotone-increasing and the search recovers the sparse generative model the data support — a state with genuine evidence yields $\Delta F < 0$ when pruned and is kept.

Greedy multi-hypothesis... (part 4)

This is the multi-state analogue of the configured BMR sign-control result of sec. 10.4, and is verified directly in the model-reduction tests.